

that of 2016, even though bitcoin's returns in 2013 were so much greater, as shown in Figure 7.16.

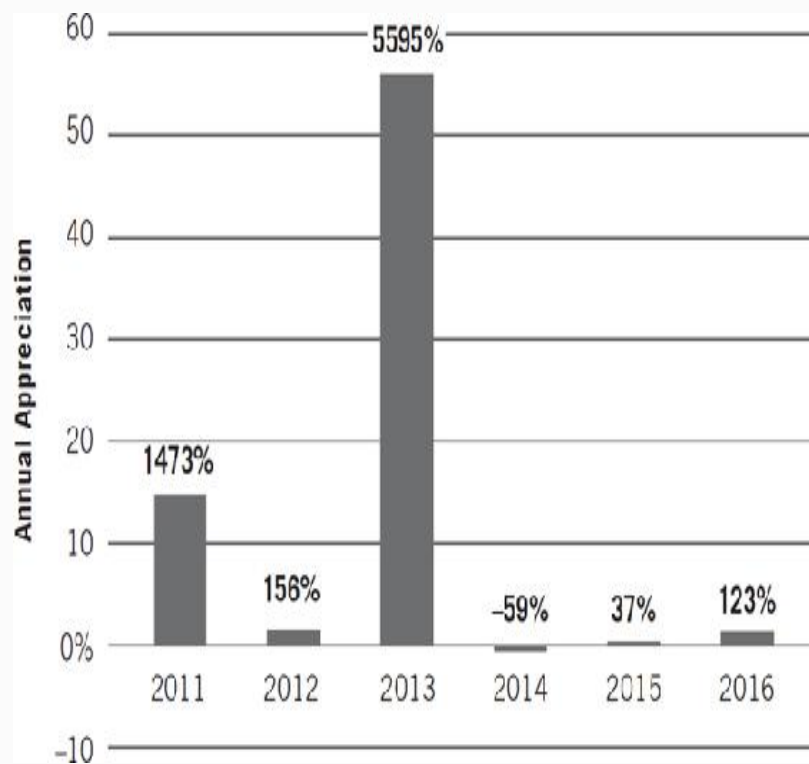


Figure 7.16 ■ Bitcoin's annual appreciation
Data sourced from CoinDesk

With capital appreciation in 2013 at 45 times greater than that of 2016, it would be reasonable to expect bitcoin in 2013 to have had a Sharpe ratio many times greater than in 2016. However, this is where both daily volatility and the way the Sharpe ratio is calculated come into play.¹¹ First, volatility in 2013 was triple that of 2016, which implies investors were taking three times as much risk in 2013 as in 2016. This allowed 2016 to have much lower returns but still have a risk-reward ratio within the same ballpark as 2013. Second, the